

Experiment No: 02

Experiment Name: Performing file and directory management and understanding file permissions in Linux.

Theory:

Linux provides a command-line interface for efficient file and directory management. Commands such as `pwd` display the current working directory, `ls` lists files and directories, `mkdir` creates directories, `cd` changes the current directory, `touch` creates empty files, `cp` copies files, `mv` moves or renames files, and `rm` or `rmdir` removes files or directories. These commands enable users to organize and manage data effectively.

The `ls -l` command displays detailed information about files and directories in a long listing format. A typical output such as `-rwxr-xr-- 1 user group 1024 Jul 15 10:30 file.txt` consists of several fields: the first character indicates the file type (- for a regular file, d for a directory, l for a symbolic link); the next nine characters represent permissions divided into Owner (rwx), Group (r-x), and Others (r--). The following fields show the number of hard links, owner name, group name, file size (bytes), last modification date and time, and finally the file or directory name.

Command	Working Principle / Description
<code>pwd</code>	Displays the absolute path of the current working directory.
<code>ls -a</code>	Lists the all visible & hidden files/folder within the directory.
<code>ls -l</code>	list the file and folder with some details ➔ <code>.rw-r--r-- 0 user 2 Aug 19:45 file1.txt</code> ➔ <code>drwxr-xr-x - user 2 Aug 19:46 fol1</code>
<code>mkdir [folder name] ... [folder name]</code>	Creates multiple directories with the specified name.
<code>mkdir -p [folder name]/ ... /[folder name]</code>	Creates nested directories.
<code>touch [file name] ... [file name]</code>	Creates multiple empty files.
<code>cat [file name]</code>	Reads and outputs the contents of a file directly to the terminal.
<code>nano [file name]</code>	Opens a file inside the Nano terminal text editor for editing.
<code>man</code>	to know the existence commands

ScreenShot From Terminal:

```
> pwd
/home/tingtong

> ls
dwxr-xr-x - tingtong 27 Jul 09:08 Desktop
dwxr-xr-x - tingtong 26 Jul 12:55 Documents
dwxr-xr-x - tingtong 2 Aug 19:29 Downloads
dwxr-xr-x - tingtong 26 Jul 23:15 lab1_200726
dwxr-xr-x - tingtong 25 Jul 12:47 Music
dwxr-xr-x - tingtong 27 Jul 01:38 'My Pictures'
dwxr-xr-x - tingtong 26 Jul 23:17 Pictures
dwxr-xr-x - tingtong 25 Jul 12:47 Projects
dwxr-xr-x - tingtong 25 Jul 12:47 Public
dwxr-xr-x - tingtong 25 Jul 12:47 Templates
dwxr-xr-x - tingtong 30 Jul 00:58 testfolder
dwxr-xr-x - tingtong 25 Jul 12:47 Videos
dwxr-xr-x - tingtong 26 Jul 12:30 'VirtualBox VMs'

> mkdir os_lab2

> cd os_lab2

> mkdir fol1 fol2 fol3

> touch file1.txt file2.txt file3.txt

> mkdir -p fol1/a/b/c

> tree fol1
fol1
├── a
│   └── b
│       └── c

> nano file2.txt

> cp file2.txt file3.txt

> ls -l
-rw-r--r-- 0 tingtong 2 Aug 19:45 file1.txt
-rw-r--r-- 77 tingtong 2 Aug 19:55 file2.txt
-rw-r--r-- 77 tingtong 2 Aug 19:55 file3.txt
dwxr-xr-x - tingtong 2 Aug 19:46 fol1
dwxr-xr-x - tingtong 2 Aug 19:44 fol2
dwxr-xr-x - tingtong 2 Aug 19:44 fol3

> touch .hiddenfile.txt

> ls -a
-rw-r--r-- 0 tingtong 2 Aug 19:56 .hiddenfile.txt
-rw-r--r-- 0 tingtong 2 Aug 19:45 file1.txt
-rw-r--r-- 77 tingtong 2 Aug 19:55 file2.txt
-rw-r--r-- 77 tingtong 2 Aug 19:55 file3.txt
dwxr-xr-x - tingtong 2 Aug 19:46 fol1
dwxr-xr-x - tingtong 2 Aug 19:44 fol2
dwxr-xr-x - tingtong 2 Aug 19:44 fol3

> bat file3.txt
File: file3.txt
1 hi
2 this is Abdullah Al Mahfuz id 241311058
3 dept. of CSE Varendra University
```

Discussion:

This experiment demonstrated how Linux commands are used to perform essential file and directory management tasks while maintaining security through file permissions. Practical use of commands such as mkdir, touch, cp, mv, rm, and showed how files and directories can be created, organized, modified, and protected. Understanding these commands is fundamental for efficient system administration and daily Linux usage.

The ls -l command was particularly useful for verifying file properties and permissions after performing different operations. By interpreting each field of the long listing format, users can identify the file type, permission settings, ownership, file size, and modification time. This knowledge helps ensure proper access control, troubleshoot permission-related issues, and maintain a secure and well-organized Linux file system.